Half Symbol OFDM Modulation

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Abstract

Based on the symmetry characteristic of subcarriers in the traditional orthogonal frequency division multiplexing (OFDM) system, we propose a half symbol OFDM (HS-OFDM) modulation method to further improve the bandwidth efficiency of communication system, the method applicable to OFDM systems whose subcarriers are modulated by BPSK or PAM. The spectra overlap percentage of adjacent subcarriers is raised from 50% to 75%, and the bandwidth efficiency upper limit of the passband OFDM system is raised from 1Baud/Hz to 2Baud/Hz, which is twice the efficiency of the traditional OFDM system.

1. Introduction

The concept of OFDM (orthogonal frequency division multiplexing) has been proposed for decades [1], and OFDM is a multi-carrier modulation technique where data symbols modulate a parallel collection of regularly spaced sub-carriers. The sub-carriers have the minimum frequency separation required to maintain orthogonality of their corresponding time domain waveforms, yet the signal spectra corresponding to the different sub-carriers overlap 50% in frequency [2-4], and the spectral overlap results the bandwidth efficiency of OFDM system approximate to the Nyquist limit of bandpass system—1Baud/Hz.

Over the last several decades, the researches of OFDM mainly focus on other two aspects. The first is to overcome the sensitivity to time and frequency synchronization problems [5-7]. The second is to reduce the high PAPR of the OFDM signals [8-9]. Till now, researches are quite comprehensive, by its extrusive advantages; OFDM has gained popularity in a number of applications including digital television and audio broadcasting, wireless local area networks, and 4G cellular land mobile radio systems. As the spectrum resources become increasingly tense and the demands for high speed applications become more urgent, it's emergent and necessary to dig the potential bandwidth efficiency capacities of the traditional systems.

It seems the bandwidth efficiency of traditional OFDM system (1Baud/Hz) has came to an absolute limit, but we find that the bandwidth efficiency can be further improved if we utilize the symmetry characteristic of the subcarriers. In the present letter, with the symmetry characteristic, we propose a novel multicarrier modulation method called HS-OFDM, of which the subcarriers at the transmitter are not orthogonal to each other and suffer severe ICI (inter-carrier interference), but at the receiver after whole symbol reconstitution, the subcarriers are orthogonal to each other without ICI. This method raises the bandwidth efficiency upper limit of the passband system to 2Baud/Hz.

2. Half Symbol OFDM Modulation

The traditional OFDM technique adopts sine or cosine functions which are orthogonal to each other as subcarriers, the orthogonality condition is that the minimum subchannel space must be $1/T_s$ (T_s is the symbol duration). A modulated OFDM symbol $s(t)$ can be expressed as [1-2, 4, 10-11]

$$\begin{aligned} s(t) &= \sum_{m=0}^{M-1} \Re\left\{(a_m + jb_m)\exp(j2\pi f_m t)\right\} \\ &= \sum_{m=0}^{M-1} \left[a_m \cos(2\pi f_m t) - b_m \sin(2\pi f_m t)\right] \end{aligned} \tag{1}$$

where M is the total number of subcarriers, a_m , b_m are data being transmitted corresponding to the m-th cosine subcarrier and the m-th sine subcarrier, and f_m denotes the center frequency of the m-th subcarrier.

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The modulated symbol $s(t)$ can be expressed as the sum of cosine subcarriers component and sine subcarriers component, which is

$$s(t) = s_C(t) + s_S(t) \tag{2}$$

where

$$s_C(t) = \sum_{m=0}^{M-1} a_m \cos(2\pi f_m t) \tag{3}$$

indicates the cosine subcarriers component, which is the sum of all the cosine subcarriers; and

$$s_S(t) = \sum_{m=0}^{M-1} b_m \sin(2\pi f_m t) \tag{4}$$

indicates the sine subcarriers component, which is the sum of all the sine subcarriers.

As the cycles of every subcarrier is an integer during the symbol duration, every subcarrier has the symmetrical characteristic, the sine subcarriers are odd symmetry, and the cosine subcarriers are even symmetry. And obviously, the component $s_S(t)$ is odd symmetry, and $s_C(t)$ is even symmetry.

When the subcarriers are modulated by BPSK, the modulated symbol $s(t)$ can be expressed as

$$s(t) = \sum_{m=0}^{M-1} a_m \cos(2\pi f_m t) \quad t \in [0, T_s] \tag{5}$$

Obviously, $s(t)$ is even symmetry. In practical application, if the former half or the latter half symbol is transmitted, the whole symbol can be obtained based on symmetry at the receiver. When the former or latter half symbol is transmitted, a novel multicarrier modulate method is born, and we call this novel modulate method half symbol OFDM (HS-OFDM), without loss of generality, assuming the former half symbol is transmitted, the modulator structure is shown in Figure 1.

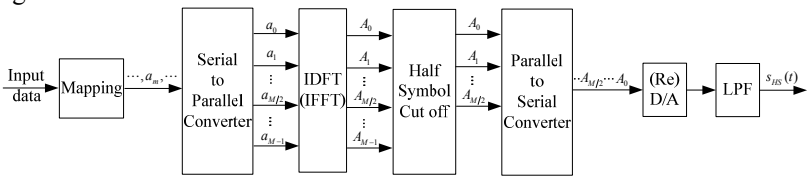


Fig.1.The modulator structure of HS-OFDM

The modulator works as follows. according to the BPSK modulation, map the coming bit data sequences to a data sequences being transmitted a_m, the serial to parallel converter will convert the serial data a_m into a group of parallel data $a_0, a_1, ..., a_{M-1}$, then perform an inverse discrete Fourier transform(IDFT) or an inverse fast Fourier transform(IFFT) to the parallel data, and the result is parallel data $A_0, A_1, ..., A_{M-1}$. Cut off the latter $M/2$ numbers and persist the former $M/2$ ones, convert the parallel data of $M/2$ numbers to serial data sequences $A_0, A_1, ..., A_{M/2-1}$. Finally, pass the real part of the serial data through a D/A converter, and a low-pass filter, then, the transmitted signal $s_{HS}(t)$ is obtained.

According to the modulation process above, a half symbol modulated signal that represents the whole OFDM symbol can be expressed as

$$s_{HS}(t) = \sum_{m=0}^{M-1} a_m \cos(2\pi f_m t) \quad t \in [0, T_s/2] \tag{6}$$

Except for the symbol duration is changed from $[0, T_s]$ to $[0, T_s/2]$, Equation (6) is the same as Equation (5). The demodulation of the half symbol modulated signal is considered in next Section.

3. Demodulation Of The Half Symbol Modulated Signal

Assume the communication system is ideally synchronized, and the interference is additive white Gaussian noise, the received signal $r_{HS}(t)$ can be expressed as

$$r_{HS}(t) = s_{HS}(t) + n(t) \quad t \in [0, T_s/2] \tag{7}$$

where $n(t)$ is a sample waveform of a zero-mean white Gaussian noise process with power spectral density of $N_0/2$.

Let the Fourier transform of $r_{HS}(t)$ is $F_{HS}(f)$, and $F_{HS}(f)$ can be expressed as

$$F_{HS}(f)=\int_{0}^{T_s/2} r_{HS}(t)e^{-j2\pi ft}dt \tag{8}$$

Then, the Fourier transform the corresponding whole OFDM symbol can be written as

$$\begin{aligned} F(f)&=\int_{0}^{T_s/2} r_{HS}(t)e^{-j2\pi ft}dt+\int_{T_s/2}^{T_s} r_{HS}(t)e^{-j2\pi ft}dt \\ &=\int_{0}^{T_s/2} r_{HS}(t)e^{-j2\pi ft}dt+\int_{T_s/2}^{T_s} r_{HS}(T_s-t)e^{-j2\pi ft}dt \end{aligned} \tag{9}$$

Adopt the even symmetry characteristic of the received symbol, coupled with the time domain properties of Fourier transform, we can get

$$F(f)=F_{HS}(f)+F_{HS}(-f) \tag{10}$$

Form the properties a real function's Fourier transform, $F(f)$ reduces to

$$F(f)=F_{HS}(f)+F_{HS}^{*}(f) \tag{11}$$

where $F_{HI}^{*}(f)$ is the conjugate of $F_{HI}(f)$. Discretize Equation (11), we can get the demodulate result of the half symbol OFDM modulated signal, more details will not be given here.

Form Equation (11), it is clear that half symbol OFDM transmission is equivalent to whole symbol OFDM transmission when each subcarrier employs BPSK modulation. With further derivation, we can get the error probability of the HS-OFDM transmit system, which is given by

$$\begin{aligned} P_b&=\frac{1}{\sqrt{2\pi}}\int_{\sqrt{2E_b/N_0}}^{\infty}\exp(-\frac{t^2}{2})dt \\ &=Q(\sqrt{\frac{2E_b}{N_0}}) \end{aligned} \tag{12}$$

where E_b denotes the average bit energy of the half symbol modulated signal, and Q indicate the Q function.

4. Bandwidth Efficiency

For traditional OFDM system, if the symbol[12] duration is T_s, the main lobe of each subcarrier will be $2/T_s$, the frequency space of adjacent subcarriers will be $1/T_s$, and the spectra overlapped between adjacent subcarriers will be $1/T_s$ too, which means that the spectra overlap percentage is 50%. The number of subcarriers, m, the subcarrier's bandwidth, B_0 ($B_0=2/T_s$),and the total transmission bandwidth, B_{OFDM}, satisfies

$$B_{OFDM}=(m+1)B_0/2 \tag{13}$$

Because every subcarrier represent 1Baud, so the Baud transmission rate of the traditional OFDM system is $R_{OFDM}=m/T_s=mB_0/2$, and the bandwidth efficiency will be

$$\eta_{OFDM}=\frac{R_{OFDM}}{B_{OFDM}}=\frac{m}{m+1}\quad \text{(Baud/Hz)} \tag{14}$$

Obviously, as the number of subcarriers *m* increases, the bandwidth efficiency of the traditional OFDM system approach to 1Baud/Hz.

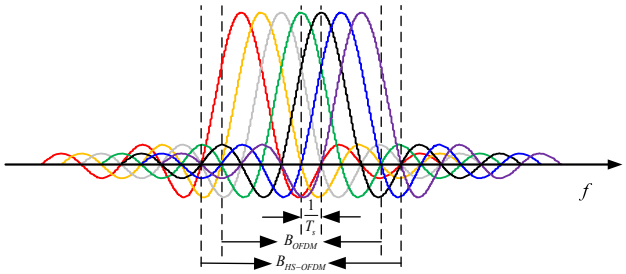


Fig.2. Spectra of subcarriers of the HS-OFDM system

As for the half symbol OFDM system, according to the relationship between time duration and bandwidth, the main lobe of each subcarrier which is twice as much as the original will be $2B_0$, but the center frequency and the frequency

space of each subcarrier will not change. So, the spectra overlapped between adjacent subcarriers will raised from $1/T_s$ to $3/T_s$, and the spectra overlap percentage will be

$$\delta = \frac{3/T_s}{4/T_s} = 75\% \tag{15}$$

The spectra of subcarriers of the HS-OFDM system are shown in Figure 2.

Form Figure 2, we can get that the number of subcarriers m, the subcarrier's bandwidth $2B_0$, and the total transmission bandwidth $B_{HS-OFDM}$ satisfies

$$B_{HS-OFDM} = (m+3) \cdot \frac{B_0}{2} \tag{16}$$

As the symbol rate of HS-OFDM system is twice as fast as the traditional OFDM system, we can get $R_{HS-OFDM} = 2R_{OFDM} = mB_0$, and the bandwidth efficiency of HS-OFDM system will be

$$\eta_{HS-OFDM} = \frac{R_{HS-OFDM}}{B_{HS-OFDM}} = \frac{2m}{m+3} \tag{17}$$

The upper limit of $\eta_{HS-OFDM}$ is 2Baud/Hz, which is twice the efficiency of the traditional OFDM system and the Nyquist bandwidth efficiency limit. To obtain the Nyquist bandwidth efficiency limit (1Baud/Hz), the HS-OFDM system only needs 3 subcarriers.

5. Conclusion

In this paper, we considered a half symbol OFDM modulation method based on the symbol symmetry characteristic when the subcarriers are modulated by BPSK. In particular, the subcarriers transmitted are not orthogonal to each other, and there exists severe ISI, the transmit condition doesn't satisfies the Nyquist criterion at all, but at the receiver after whole symbol reconstitution, the subcarriers are orthogonal without ISI. The error performance is same as the convention OFDM system, but the bandwidth efficiency is twice the value of the convention system.

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